A3.1 Diversity of organisms (HL)

NAME	DATE	TOTAL
		_____ / 20

About 20 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 MARKS

1 STATE the biological species concept. [A3.1.12] [1]

2 DEFINE horizontal gene transfer. [A3.1.12] [1]

3 STATE the chromosome number (2n) of a mule. [A3.1.13] [1]

4 DEFINE DNA barcode. [A3.1.15] [1]

5 STATE what is meant by a couplet in a dichotomous key. [A3.1.14] [1]

Section B: Understanding

7 MARKS

6 EXPLAIN why the biological species concept is hard to apply to organisms that reproduce asexually. [A3.1.12] [2]

7 EXPLAIN why a hybrid of two species with different chromosome numbers is usually infertile. [A3.1.13] [3]

8 OUTLINE two advantages of eDNA surveys over catching and counting organisms. [A3.1.15] [2]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice.

9 [A3.1.14]

The table describes the leaves of five trees.

TREE	LEAF DIVIDED INTO LEAFLETS?	LEAF EDGE	LEAVES ARRANGED
P	Yes	Toothed	Opposite
Q	Yes	Smooth	Alternate
R	No	Lobed	Alternate
S	No	Toothed	Alternate
T	No	Toothed	Opposite

CONSTRUCT a dichotomous key to identify the five trees. [4]

10 [A3.1.15]

Water from three ponds was tested for eDNA. The table shows the number of DNA sequences matched to each species.

SPECIES	POND A	POND B	POND C
Great crested newt	340	0	12
Three-spined stickleback	910	1520	0
Signal crayfish (invasive)	0	205	0

a **IDENTIFY** the pond where the invasive species is present. [1]

b **EXPLAIN** why the number of sequences does not show how many animals live in each pond. [2]

c **SUGGEST** why newt DNA was found in pond C, although no newts were seen there. [1]